

Mid Point Progress Update:

Understanding Climate-Related Educational Disruptions in Pakistan Through the Lens of Data, Inequality, and Policy

PI: Dr. Zubair Khalid | Co-PI: Dr. Jessica Albrent & Dr. Mansoor Khan | Team Members: Aymen Asif , Asfra Rizwan, Hajra Javed, Mina Arif, Mahnoor Naveed and Nausherwan Malik

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Organisation: LUMS

Executive Summary

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This study examines how climate-related disruptions (floods, heatwaves, and smog) affect education in Pakistan, with particular attention to data gaps, socioeconomic inequalities, and policy responses needed to build climate resilience in the education sector. Using a systematic scoping review, key informant interviews, and secondary analysis of climate and education datasets, the research maps existing evidence, identifies what data and gaps exist across climate, education, and socioeconomic indicators, and examines how climate events differentially impact educational access and outcomes across socioeconomic groups. Findings are intended to inform evidence-based policymaking at federal and provincial levels, strengthening climate-resilient education planning and improving the integration of climate and education data systems.

Changes Since Inception Report Approval

✓ No major changes to the study objectives, research questions, or methodology.



Progress Update

Progress Against Approved Work Plan

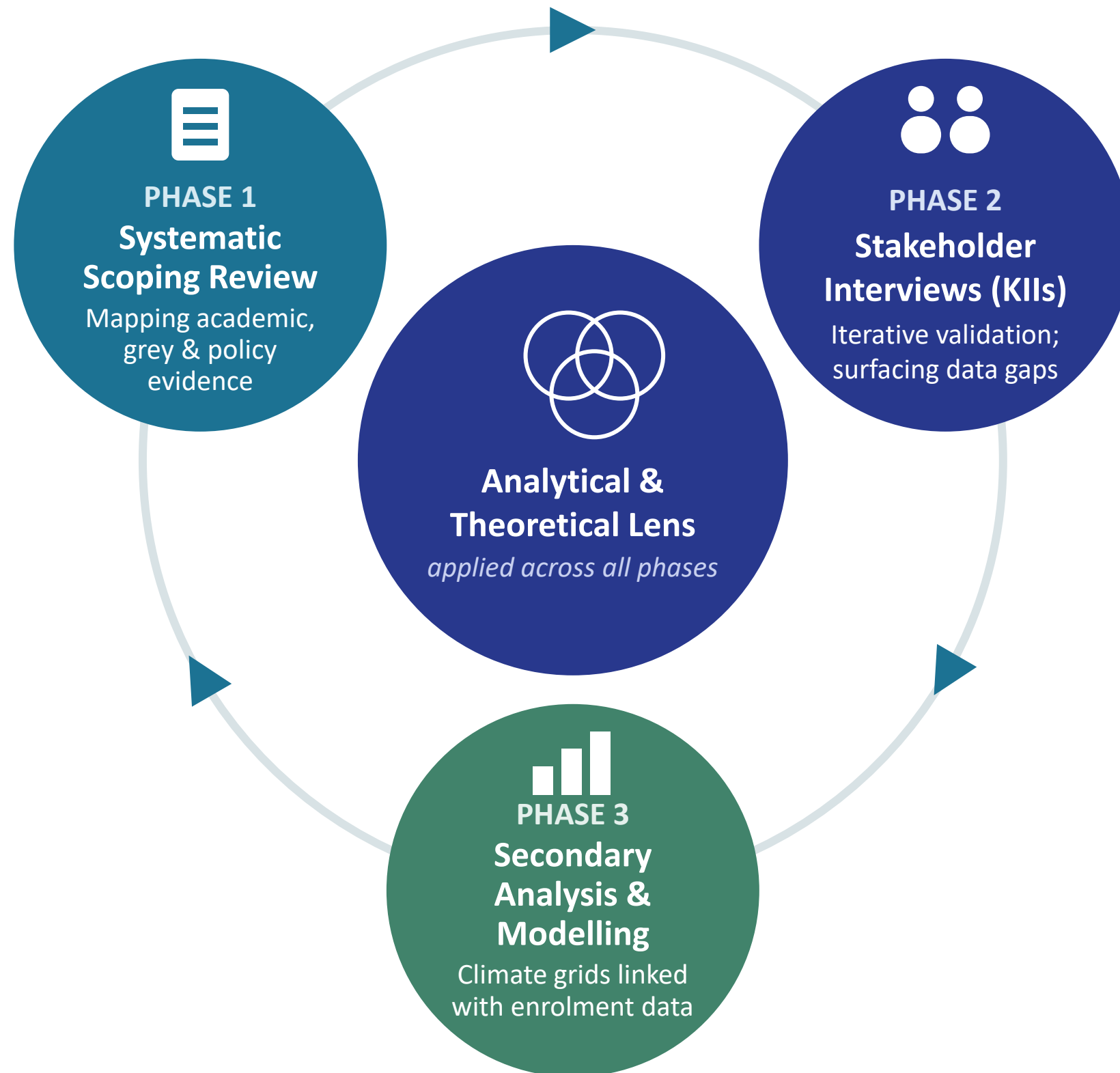
- **Scoping Review (85% Complete):** Literature identification, screening, and filtering substantially completed; thematic synthesis and evidence mapping to commence next week.
- **Key Informant Interviews (70% Complete):** Majority of planned KIIs conducted with policymakers, researchers, and data custodians using snowball sampling; 12 interviews have been completed; remaining interviews to be completed within the next **two weeks**.
- **Secondary Data Collection:** Climate datasets are currently being collected. Education datasets have largely been compiled from open sources, with a few key variables pending from the **Pakistan Institute of Education (PIE)**.

Key Challenges & Mitigation

- **Stakeholder Engagement:** Securing interviews across provinces required extensive follow-up and coordination with senior officials.
Mitigation: Leveraged professional networks, referrals, and snowball sampling to improve participation and maintain progress.
- **Administrative Data Access:** Delays in receiving selected education variables from PIE.
Mitigation: Continued analysis using available datasets while formal data requests remain under follow-up, minimizing impact on the project timeline.

Overall Status: Project implementation remains on track, with major milestones progressing according to the approved work plan and no significant deviations anticipated.

Theoretical/Conceptual Frameworks



Analytical & Theoretical Lens

- Intersectional Vulnerability Framework**
How gender, SES, and location interact to shape educational risk during climate shocks.
- Multi-Source Data Architecture**
Triangulating satellite-derived climate indices with administrative education datasets (EMIS / SIS).
- IPCC & GADRRRES CSS Integration**
Aligning recommendations with global climate-risk and Comprehensive School Safety frameworks.

Integrated Data Pillars for Analysis

- Climate Pillar** Rainfall intensity, heatwaves, dry spells, flood-exposure mapping
ERA5 · CHIRPS · Sentinel-1 SAR · 1 km daily grids
- Education Pillar** Enrolment, attendance, school locations, level & gender composition
SIS · PIE · Provincial EMIS
- SES Pillar** Socioeconomic status, gender, geographic vulnerability
Household & district datasets · 1 km Lahore SES map

KIs iterate across all phases — each phase informs the next.

Data Collection Strategies

Multi-Method Data Collection Approach

Research Component	Data Collection Strategy	Current Status
 Scoping Review	Systematic literature search using Arksey & O'Malley (2005) and PRISMA-ScR principles across academic databases and grey literature.	85% completed
 Key Informant Interviews (KIIs)	Semi-structured interviews with policymakers, researchers, development partners, and data custodians using snowball sampling.	70% completed
 Secondary Data Collection	Compilation of climate and education administrative datasets from government departments and open-access sources.	Progress underway

Key Data Sources



Academic Databases

Google Scholar, Claude AI and Consensus databases to collect data



Grey Literature

Government Reports, Blogs , European Union Reports



Development Partners

UNICEF, World Bank, ADB,



Administrative Datasets

Climate and education databases



Stakeholders

Federal & Provincial Education Departments, Disaster Management Authorities, Researchers

Data Analysis Strategies

Integrated Evidence Analysis



Scoping Review

- ✓ Six-stage scoping review framework (Arksey & O'Malley; Peters et al.)
- ✓ Three-stage screening:
 - Title screening
 - Abstract screening
 - Full-text screening
- ✓ Thematic evidence synthesis and gap mapping



Key Informant Interviews

- ✓ Thematic coding of interview transcripts
- ✓ Stakeholder validation of emerging findings
- ✓ Iterative refinement of evidence



Secondary Data Analysis

- ✓ Descriptive analysis of climate–education indicators
- ✓ Comparative provincial analysis
- ✓ Predictive modelling (where data availability permits)

Preliminary Findings

Note: Data from KIIs and the scoping review are being collected and formal analysis has not yet begun. These findings are drawn from patterns observed during data collection, rather than empirical analysis.

Intersections of SES, Climate, and Education

- The intersection of SES, Climate, and Education is almost completely absent from both the literature and the KIIs. The literature usually assumes homogenous experiences in these intersections across regional, national, and subnational contexts and/or assumes poverty.*
- KIIs discuss climate disruptions related to public schools, public school students, and their communities, leaving out private schools (who serve diverse SES groups and an estimated 40% of the population), migrants, and those in alternative educational pathways.*

Conceptualizations of Climate Disruptions in Education

- Much of the discussions in KIIs and (to a lesser extent) the literature assumes a humanitarian and urgent disaster with climate disruptions. Development and infrastructure planning are based on these assumptions. However, events such as heatwaves and smog do not register as immediate disasters, especially within the school safety framework, and are not measured or tracked, and response to these are limited at best.*

Availability, Quality, and Types of Data

- While provinces have worked closely with organizations such as PIE to collect data under the DSF, this data is recent and protocols for sharing are undefined. Obtaining data for a secondary analysis is largely dependant on networking and connections with whomever holds the data (on several levels), political will, and time.*
- KIIs reveal that data are collected and used extensively in provincial dashboards, but these have limited climate indicators and are not integrated with early warning systems (though plans to do so are being made).*
- KIIs reveal that data are not always reliable.*
- KIIs reveal that connecting climate disruptions with student learning is almost impossible, as school-level data on student achievement is not available in any province.*

Secondary Analysis: Available Datasets

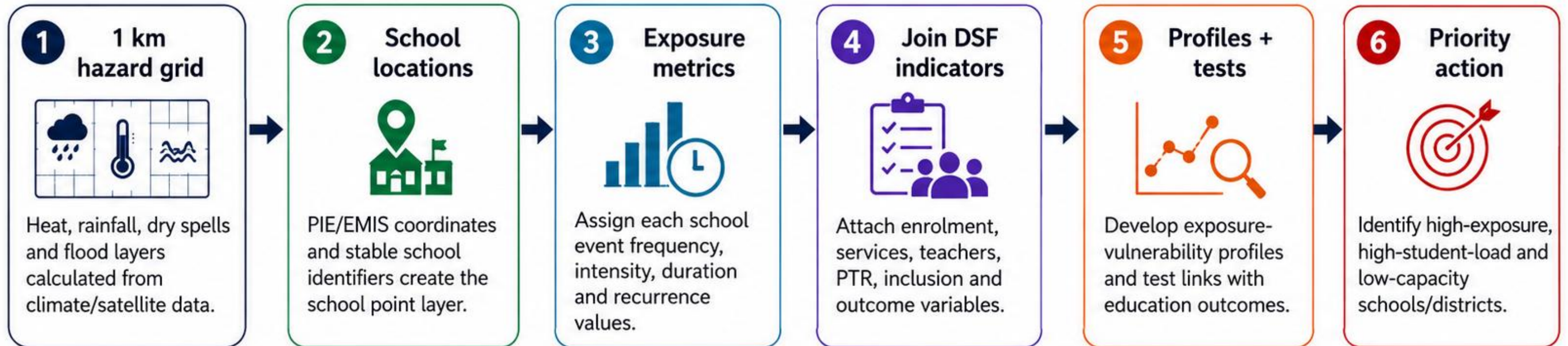
Datasets compiled for the secondary analysis of climate–education linkages in Punjab and Sindh

Dataset	Coverage	Resolution / Frequency	Role in the Analysis
Gridded climate data (temperature, rainfall, humidity, satellite data for flood mapping)	Punjab & Sindh	1 km grid; daily values	Detection of localised climate-extreme episodes (heat, extreme rainfall, dry spells)
Government school locations & characteristics	Punjab & Sindh (thousands of schools)	School-level, geo-located; level and gender type	Exposure mapping; built-in equity lens by school level and gender composition
School enrolment, Punjab	All government schools	Monthly; five-year panel	Event-study analysis of enrolment disruption around climate extremes
School enrolment, Sindh	All government schools	Single snapshot (May 2025)	Exposure headcounts; baseline for proposed periodic reporting
Socioeconomic status (SES) mapping (Lahore only)	Lahore district	1 km grid; validated in-house method	SES-integrated risk-mapping pilot (nested case study)

Datasets assembled and harmonised by the project GIS and data-science team; analysis window: 2021-2026.

Climate-Education Mapping Workflow

From climate hazard layers to school exposure, DSF integration, and priority action



Heatwave Detection and Event Mapping

Methodology: Combining wind, temperature and humidity to interpret heat wave (2021 to 2026)

1. What Is a Heatwave?



A heatwave is a prolonged period of unusually high temperature.



In this study, heat is assessed primarily using daily maximum temperature (Tmax).



Humidity and wind are also considered because they influence how hot conditions feel to people.

2. Mathematical Definition

Heatwave day if

$$T_{max}(g, d) > P_{95}(g, d)$$

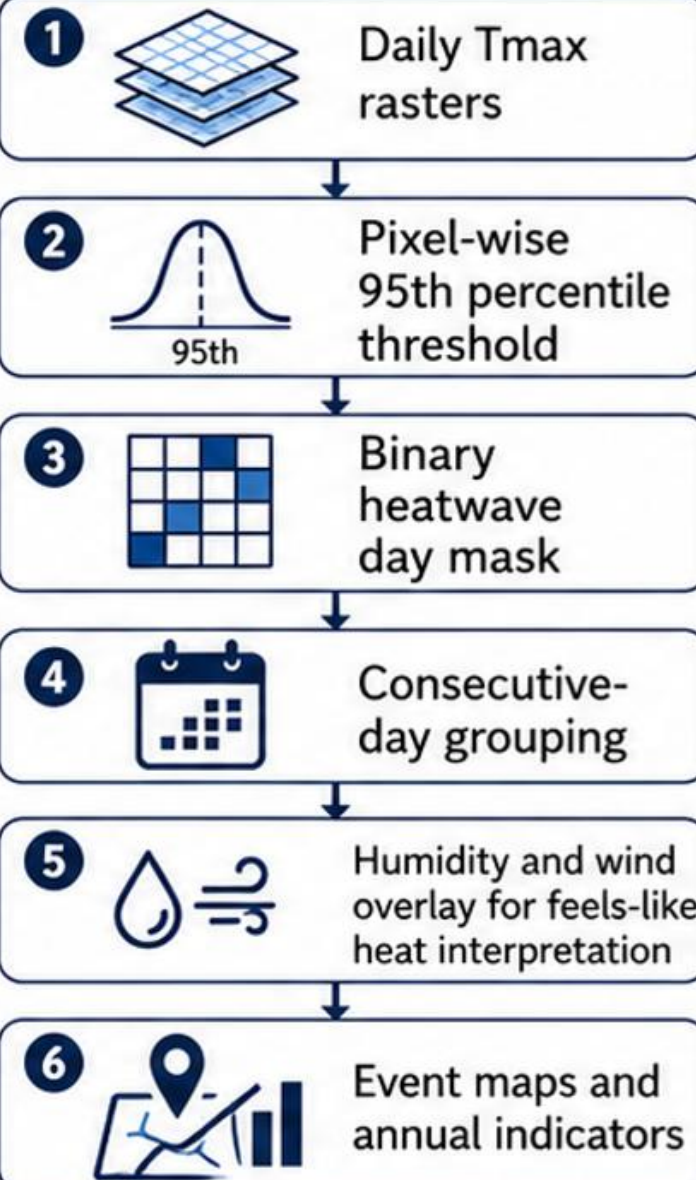
Heatwave event if consecutive heatwave days ≥ 3



Feels-like heat increases with higher humidity and lower wind speed.

Thus, humidity and wind help interpret exposure severity beyond air temperature alone.

3. Workflow



4. Key Outputs



Heatwave days



Heatwave intensity



Longest heatwave spell



Temperature variability



Event footprints and annual summaries



Outputs can later be linked to school locations and DSF indicators.

Heatwave Detection and Event Mapping

Spatial Analysis of Heatwave Events, 2021–2026

Spatial Analysis:

- Heatwave exposure; spatially uneven across Punjab and Sindh, different locations experiencing different numbers of heatwave days during the same event.
- The geographical extent and shape of the affected area changed between events; heatwaves did not follow one fixed spatial pattern.
- Longer events produced broader and more continuous zones of persistent exposure, while shorter events showed more limited or fragmented footprints.
- Locations repeatedly covered by high heatwaves across several events represent recurring heatwave-exposure zones.
- Overlaid with school data to identify which schools, districts and student populations experienced prolonged heatwave exposure.

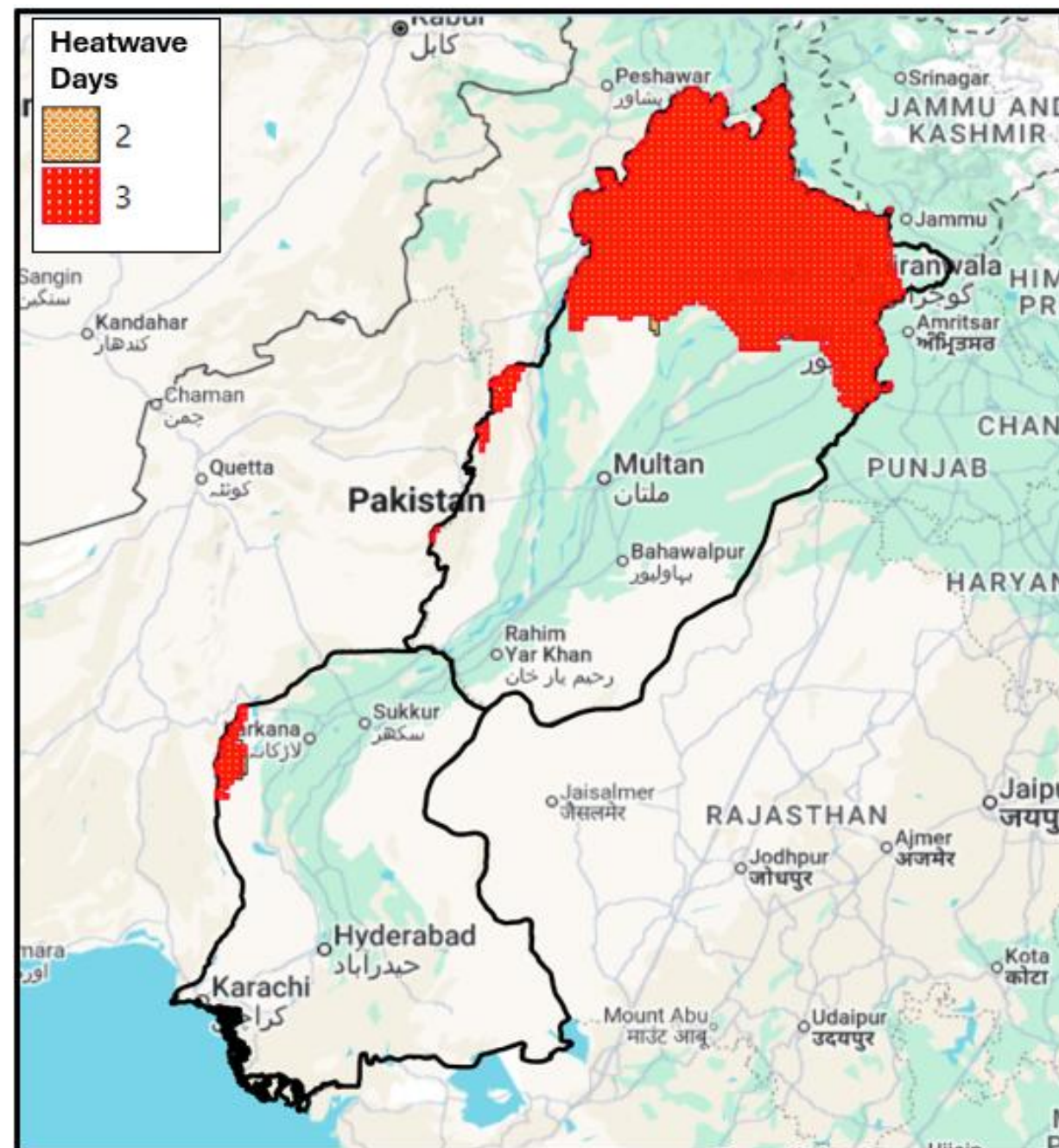
Temporal Analysis:

- Seventeen regional heatwave events were identified between 2021 and June 2026, showing repeated extreme-heat exposure across the study period.
- Event frequency varied substantially by year, ranging from 1 event in 2023 to 5 separate events in 2025.
- 2025 recorded the highest number of events, indicating repeated heatwave episodes within the same summer season.
- The longest individual heatwave occurred from 15 May to 3 June 2024, lasting 20 consecutive days, followed by the 13-day event in June 2022.
- Most events occurred between May and June, although heatwaves were also detected in April, indicating that the extreme-heat season can begin earlier than peak summer.

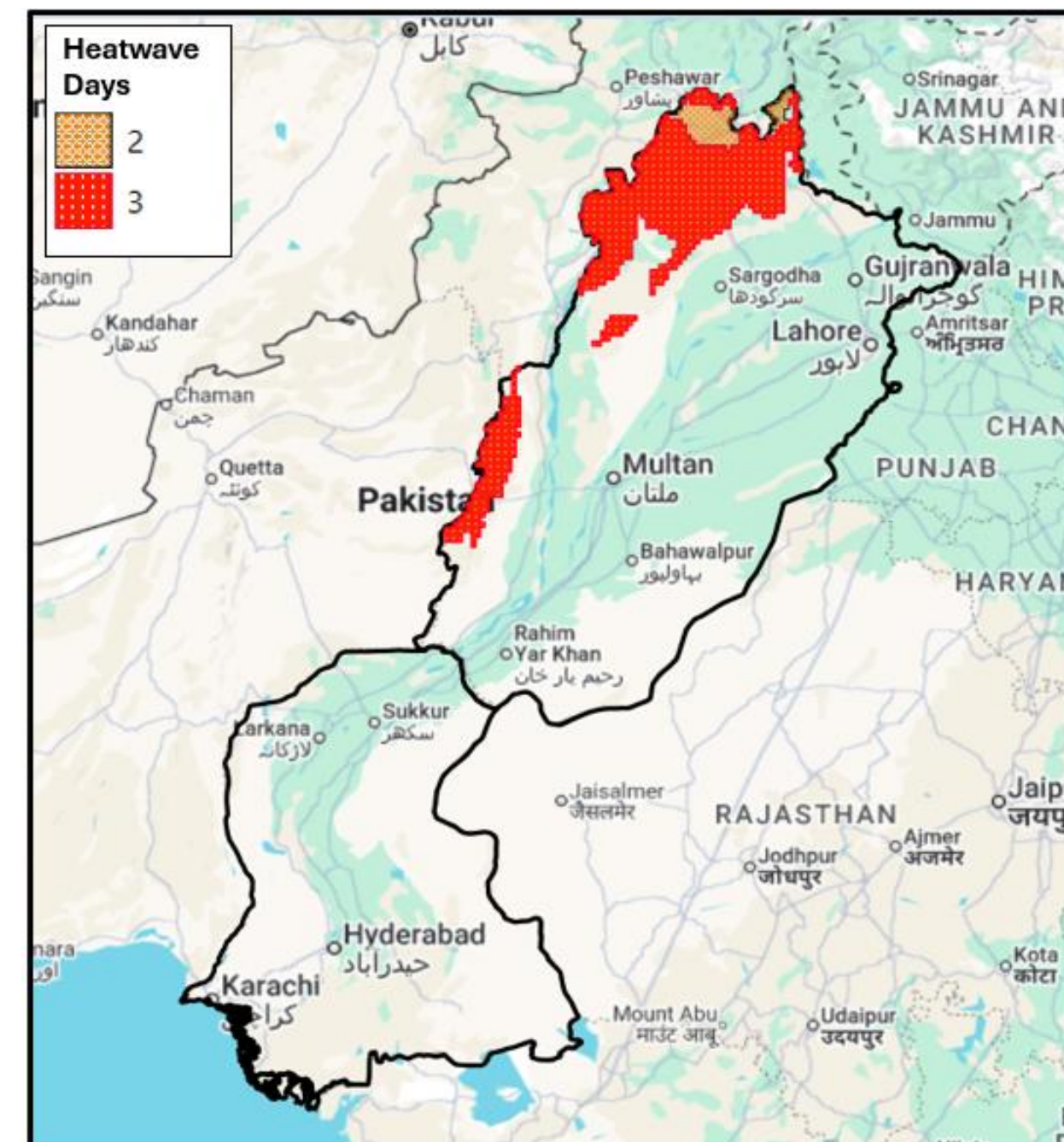
Heatwave Detection and Event Mapping

Heatwave Events, 2021

Event	Event period	Duration
2021-01	8-10 June 2021	3 days



Event	Event period	Duration
2021-02	29 June-1 July 2021	3 days



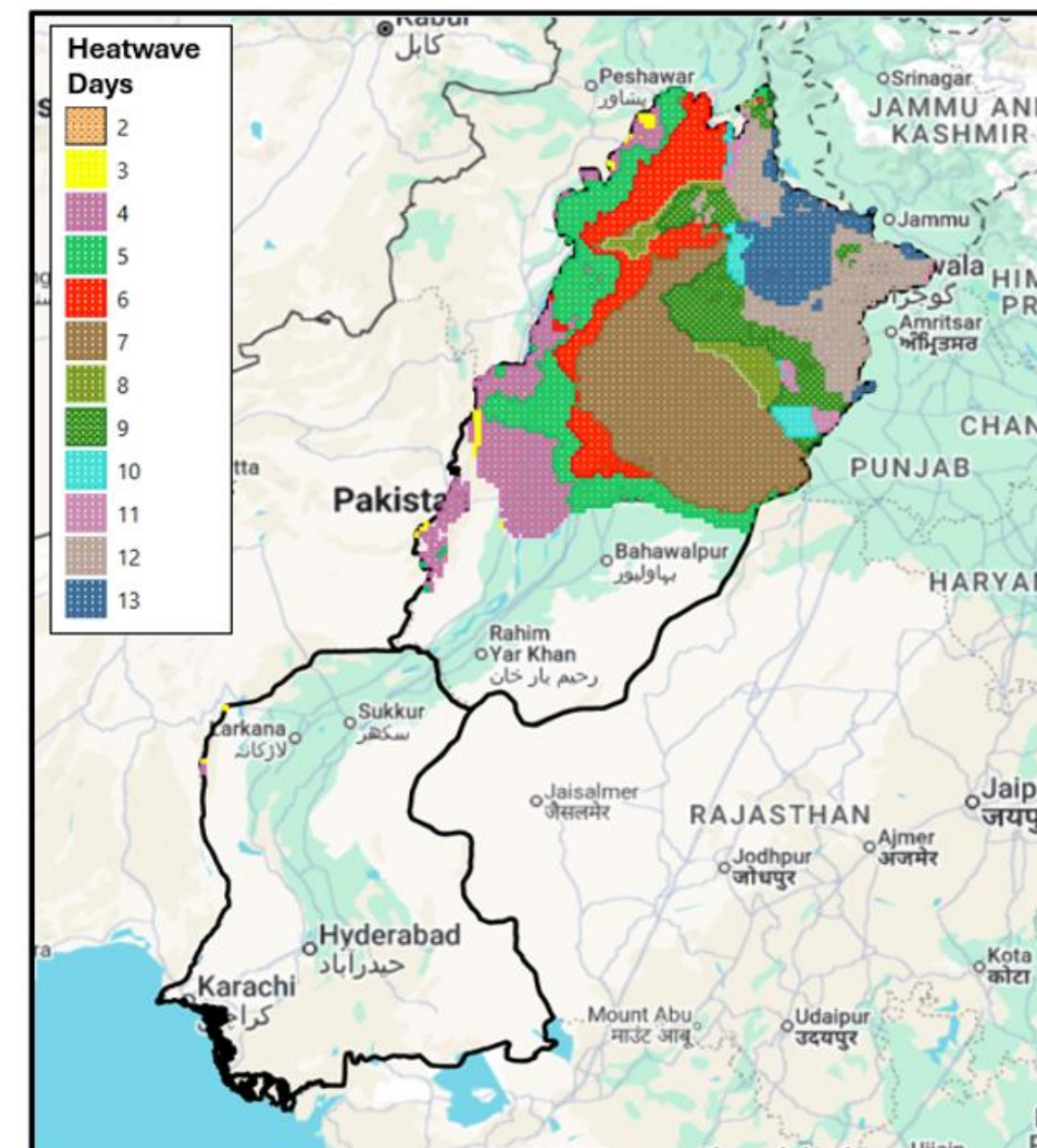
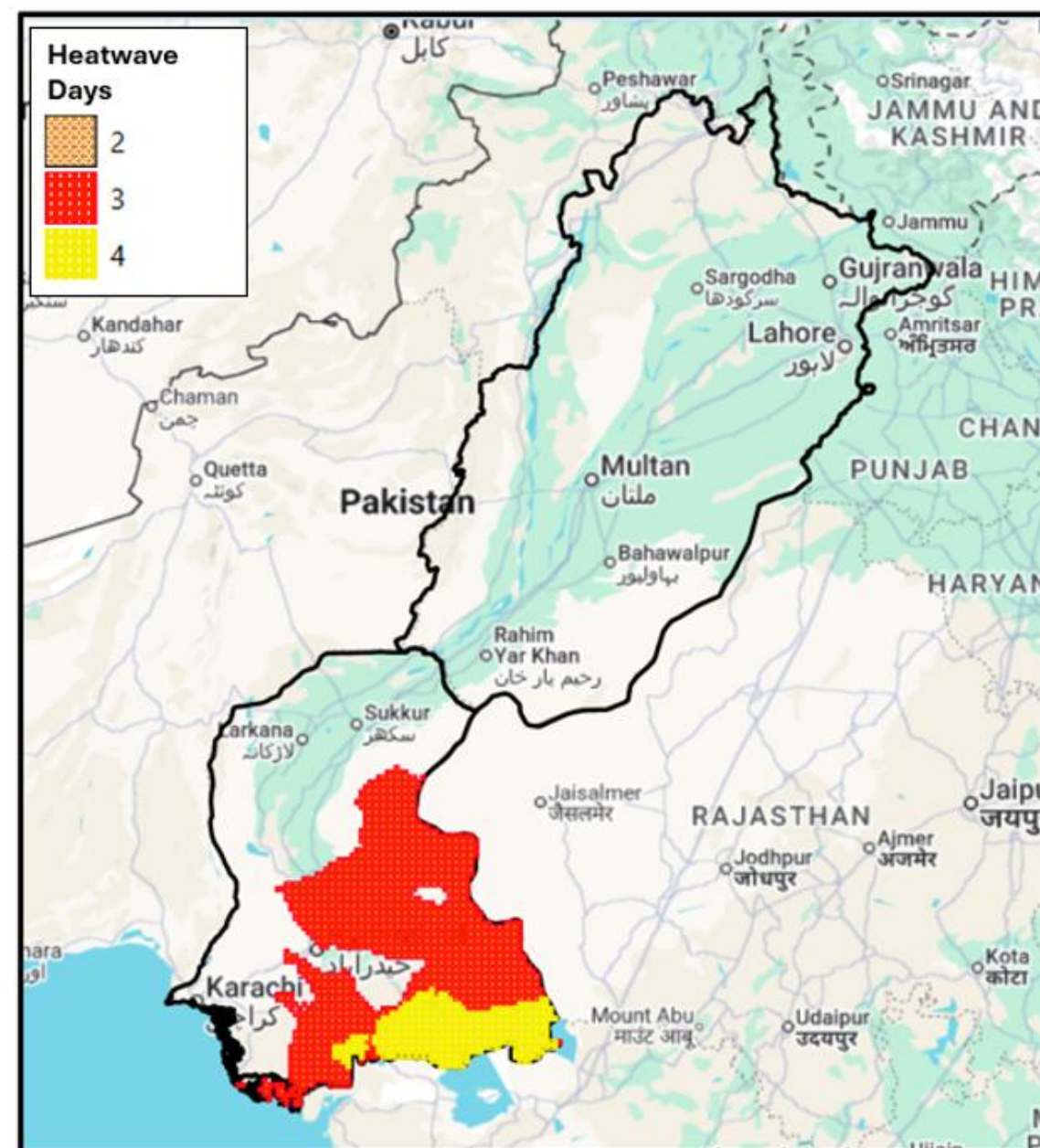
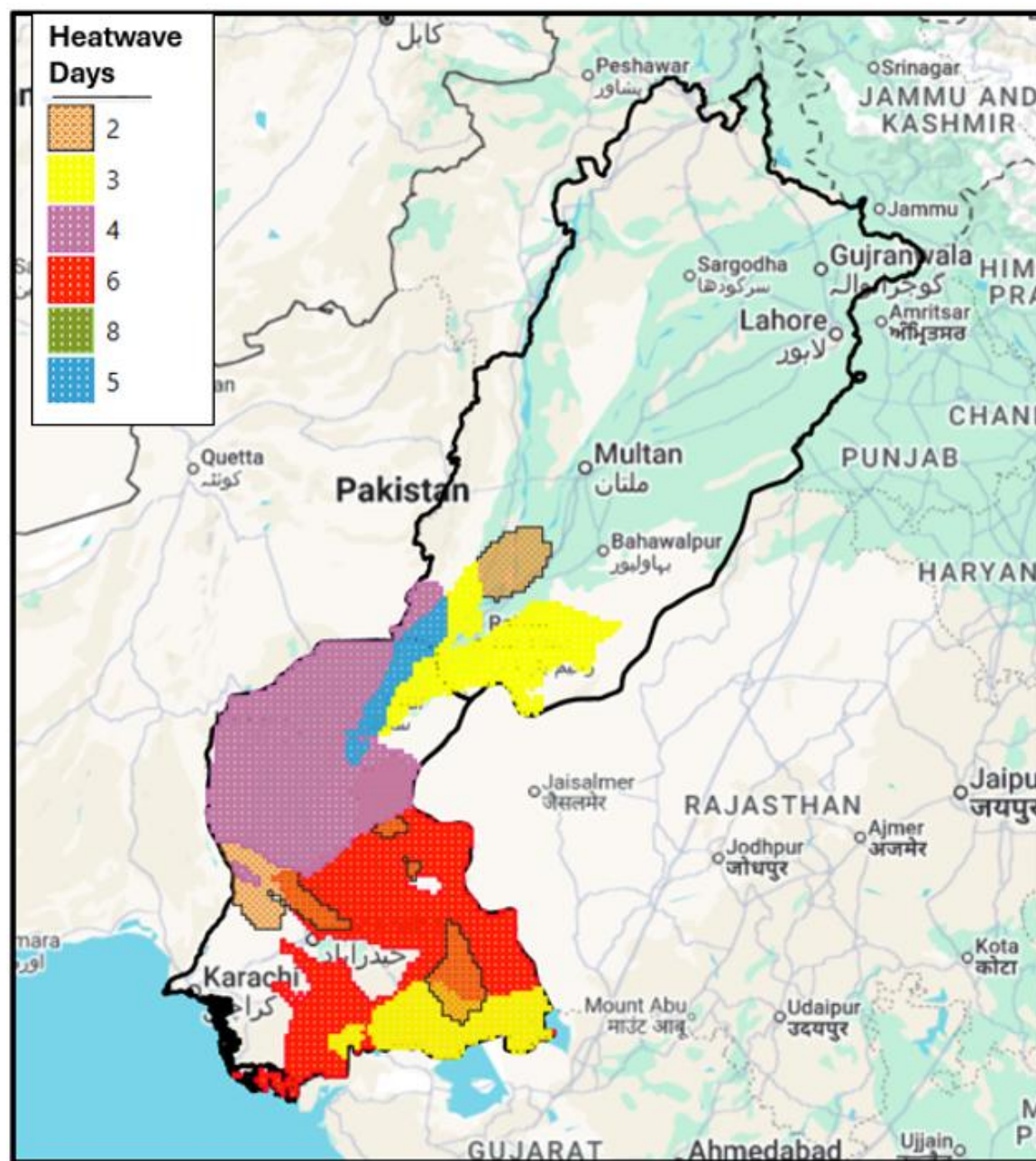
Heatwave Detection and Event Mapping

Heatwave Events, 2022

Event	Event period	Duration
2022-01	10–15 May 2022	6 days

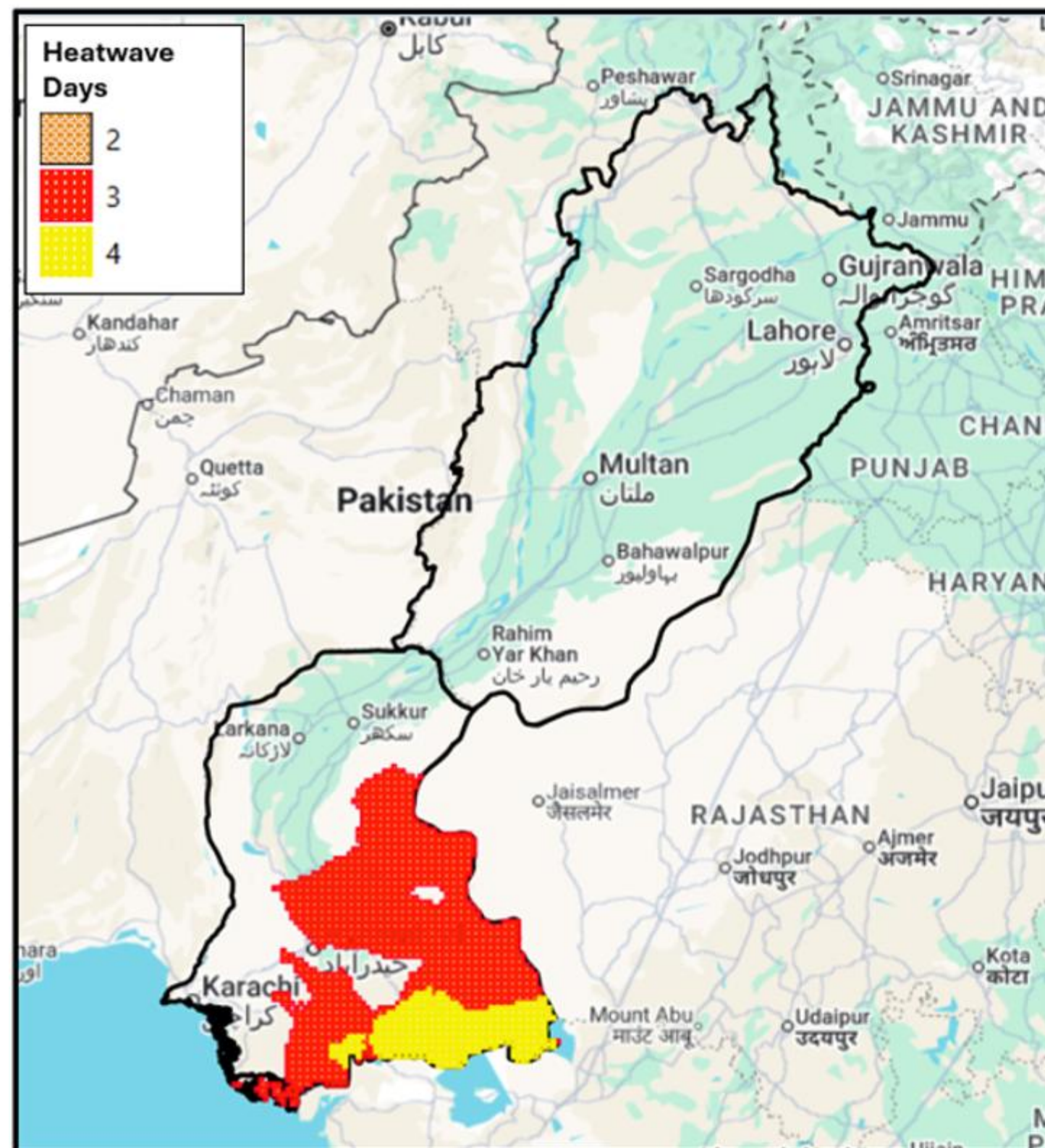
Event	Event period	Duration
2022-02	18–22 May 2022	5 days

Event	Event period	Duration
2022-03	3–15 June 2022	13 days



Heatwave Detection and Event Mapping

Heatwave Events, 2023



Event	Event period	Duration
2023-01	10-13 May 2023	4 days

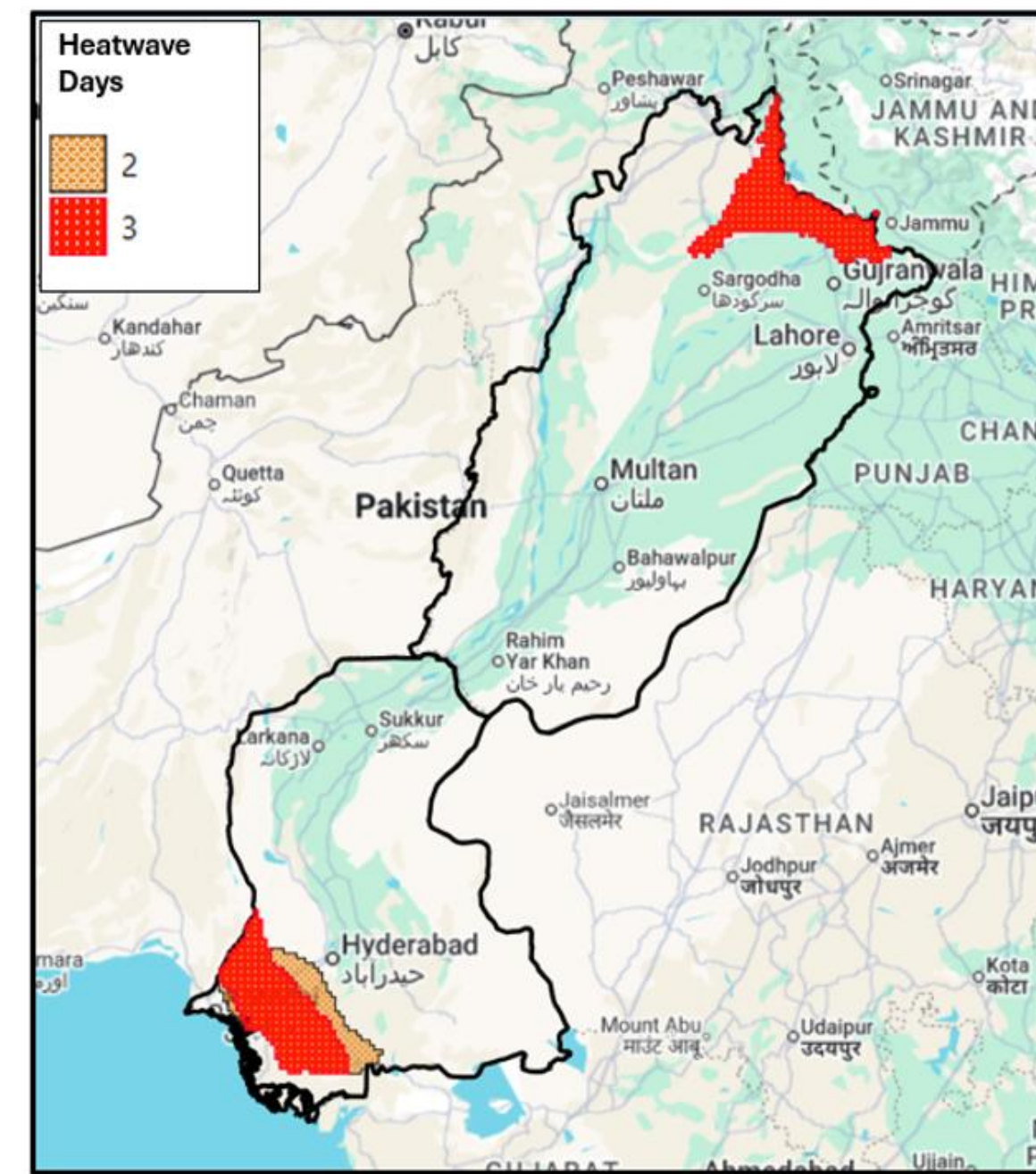
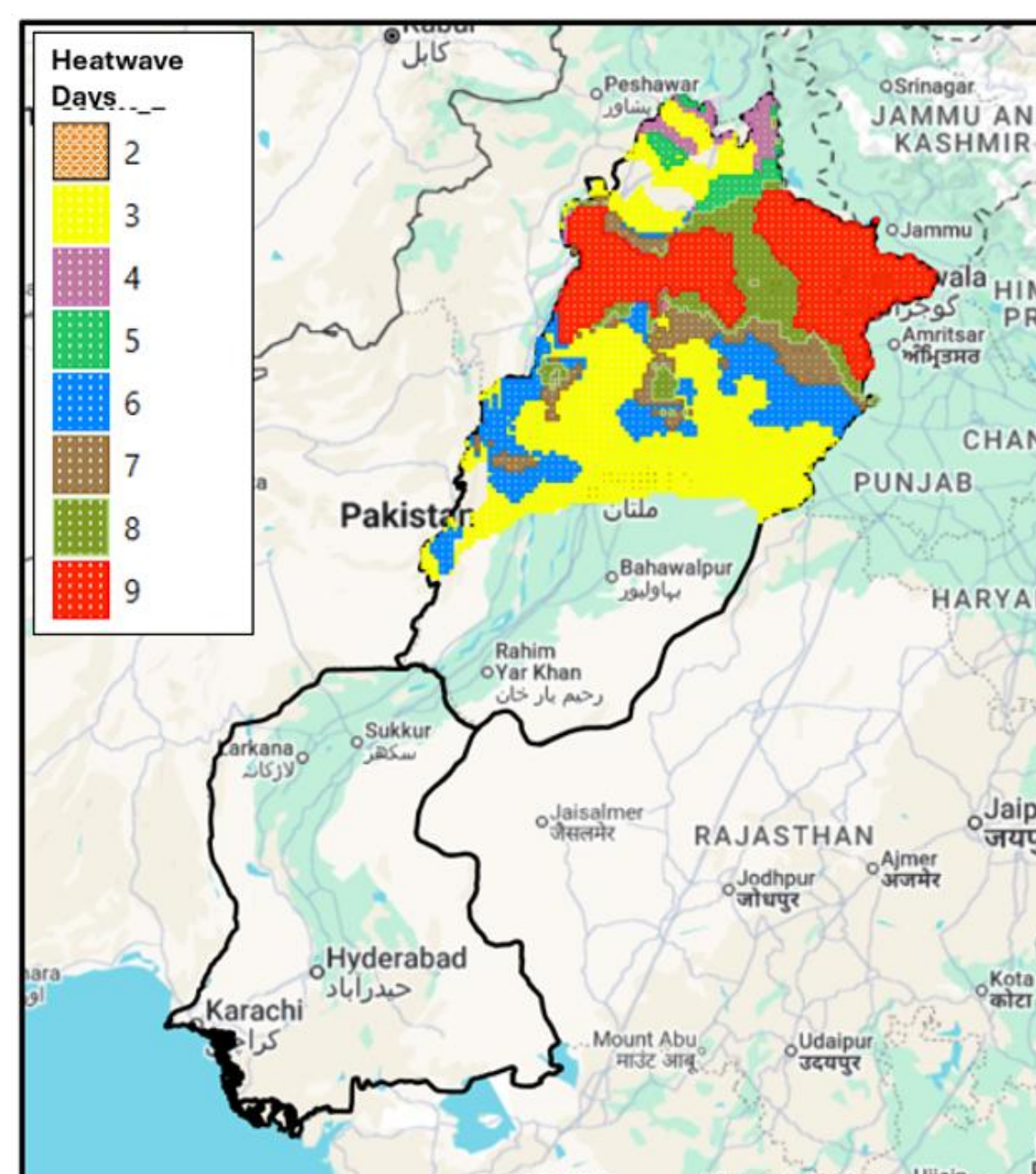
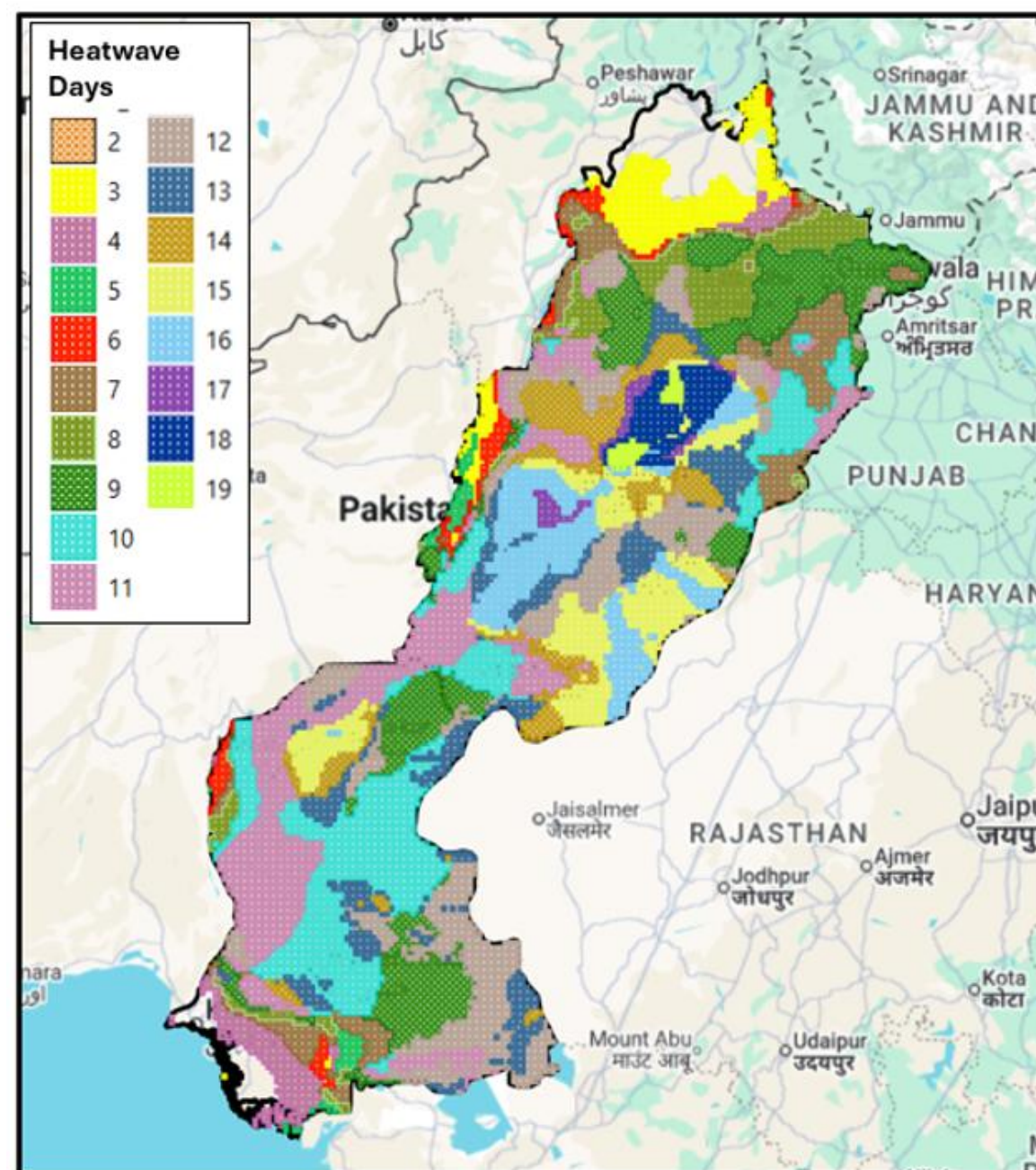
Heatwave Detection and Event Mapping

Heatwave Events, 2024

Event	Event period	Duration
2024-01	15 May–3 June 2024	20 days

Event	Event period	Duration
2024-02	11–19 June 2024	9 days

Event	Event period	Duration
2024-03	24–26 June 2024	3 days



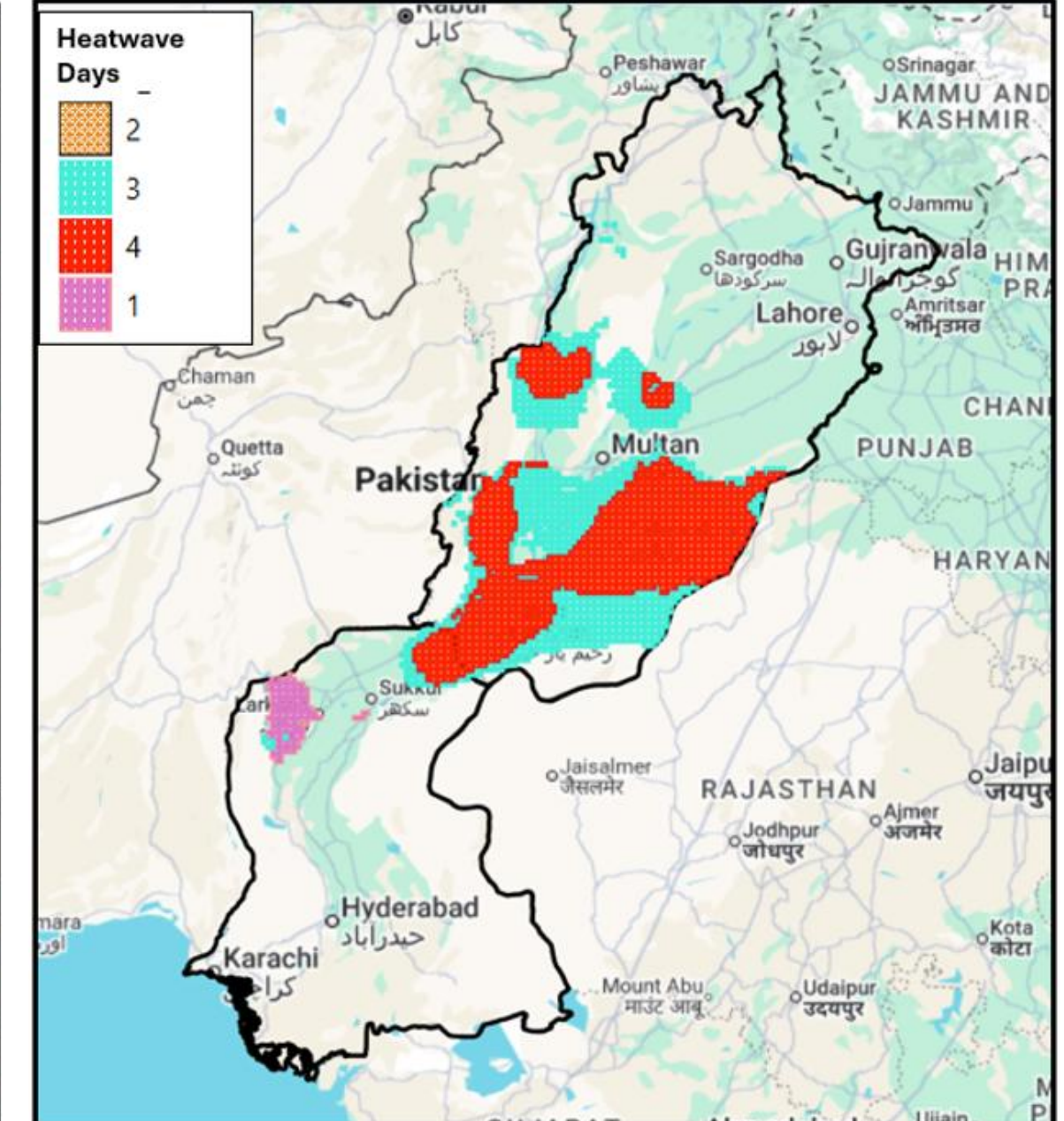
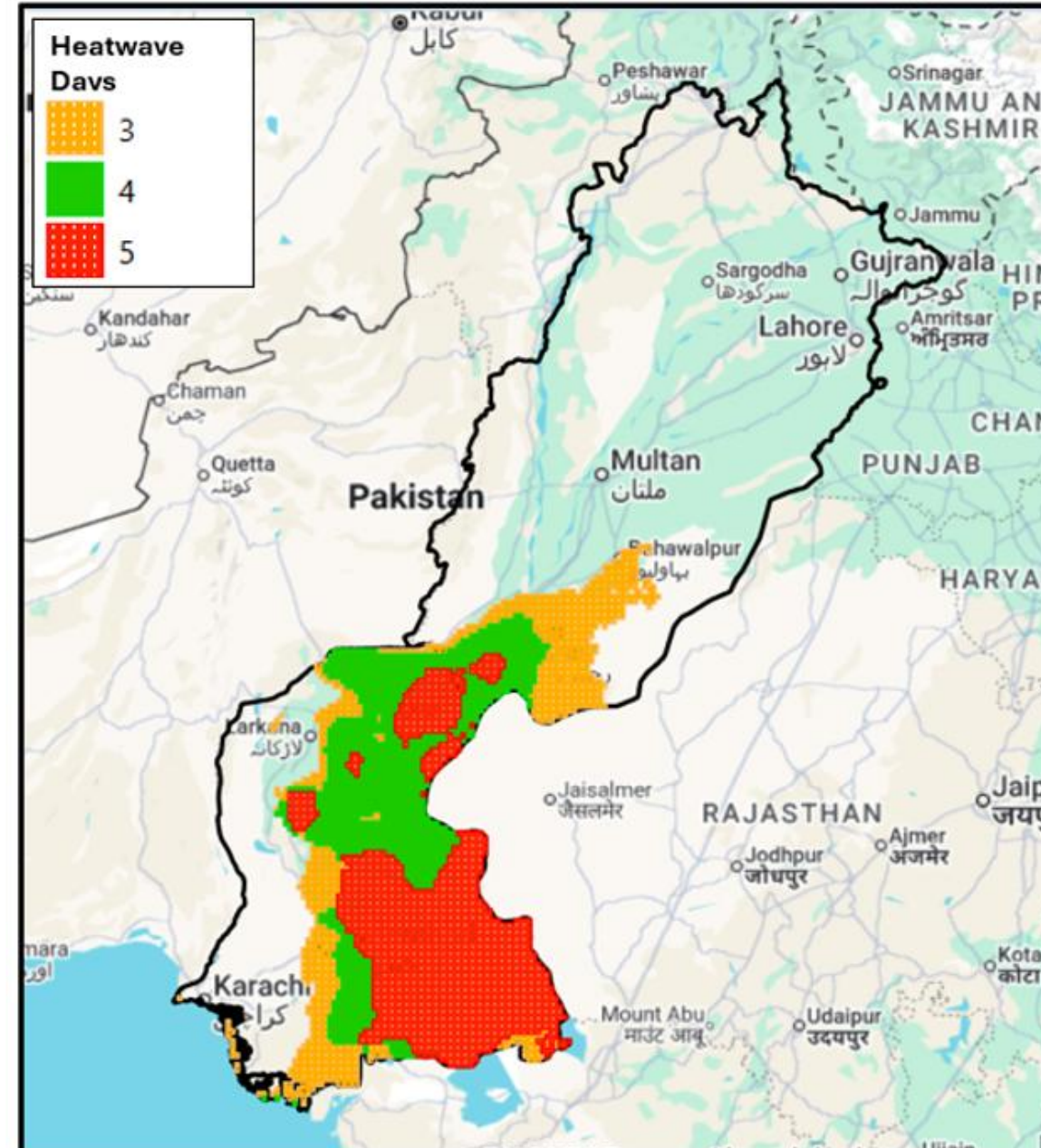
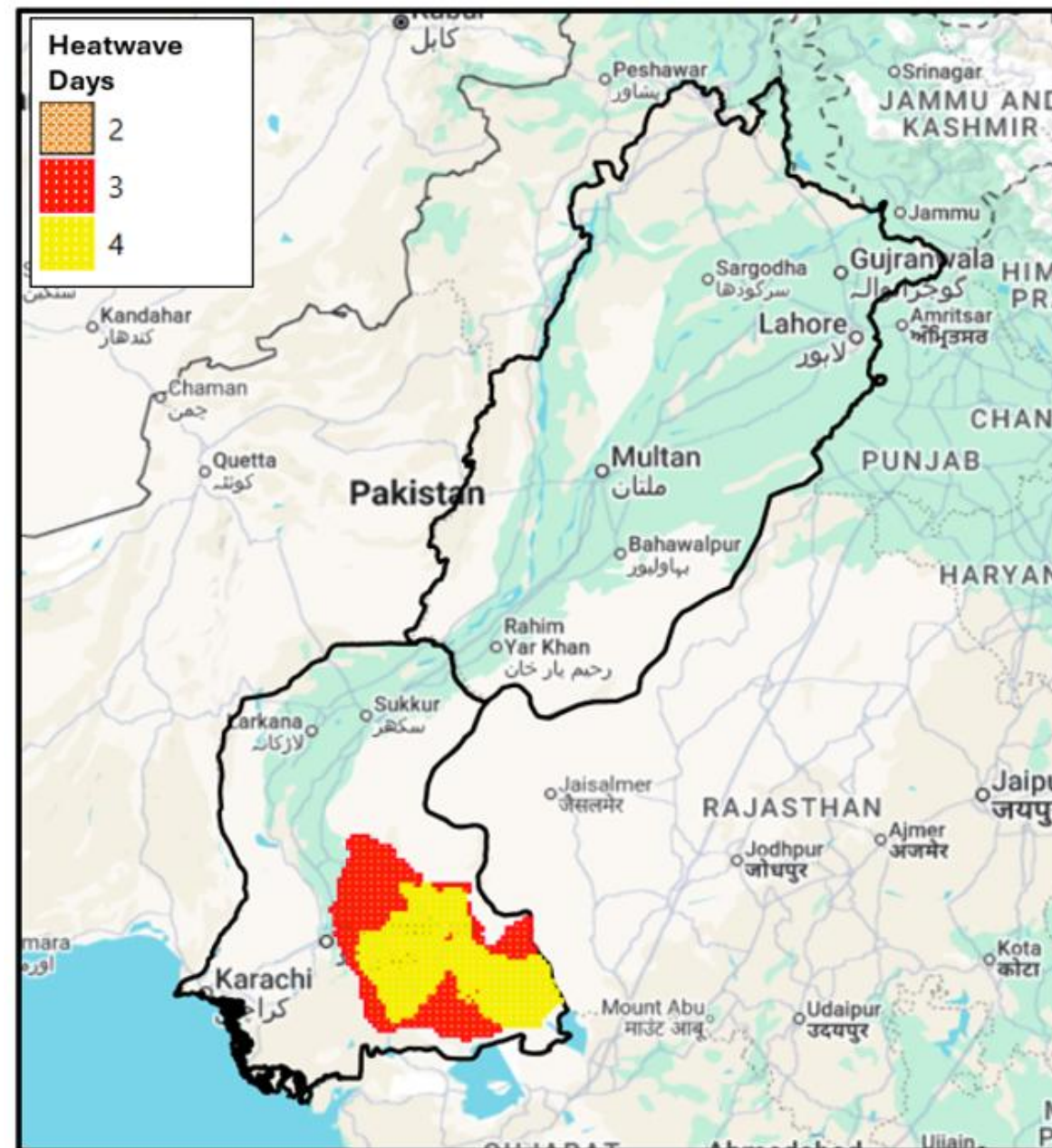
Heatwave Detection and Event Mapping

Heatwave Events, 2025

Event	Event period	Duration
2025-01	14-17 April 2025	4 days

Event	Event period	Duration
2025-02	27 April-1 May 2025	5 days

Event	Event period	Duration
2025-03	15-18 May 2025	4 days

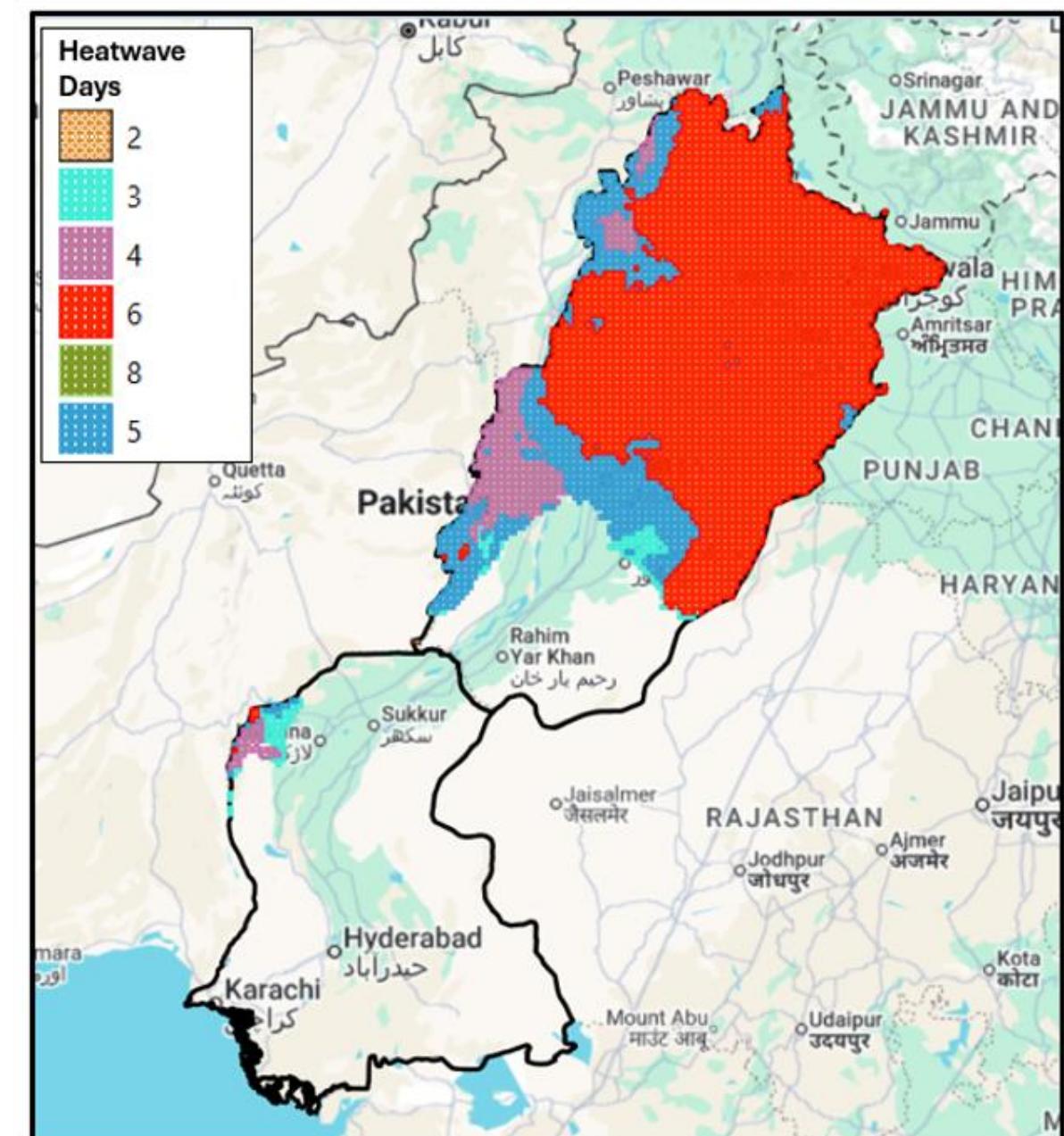
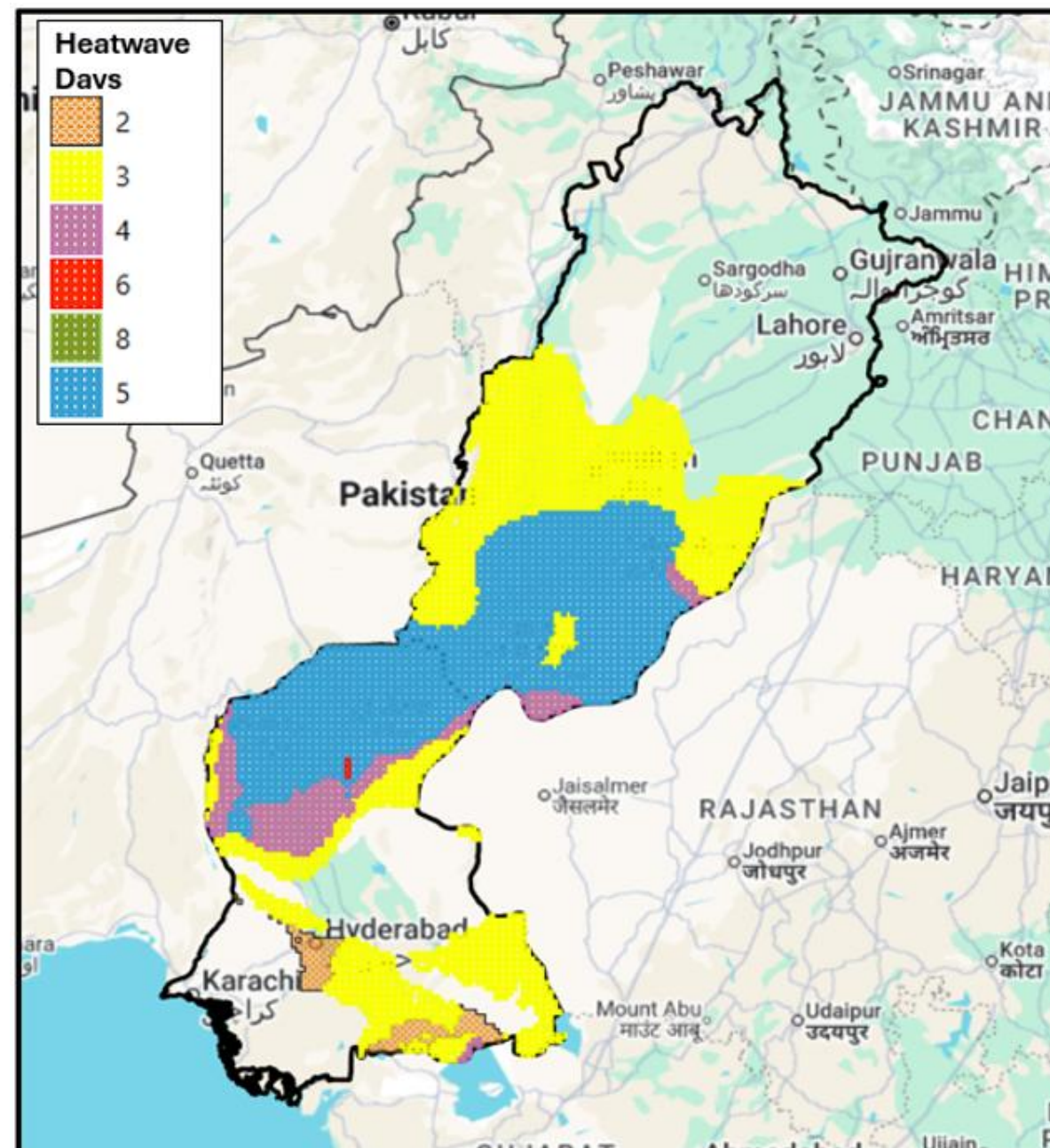


Heatwave Detection and Event Mapping

Heatwave Events, 2025

Event	Event period	Duration
2025-04	20-25 May 2025	6 days

Event	Event period	Duration
2025-05	8-13 June 2025	6 days



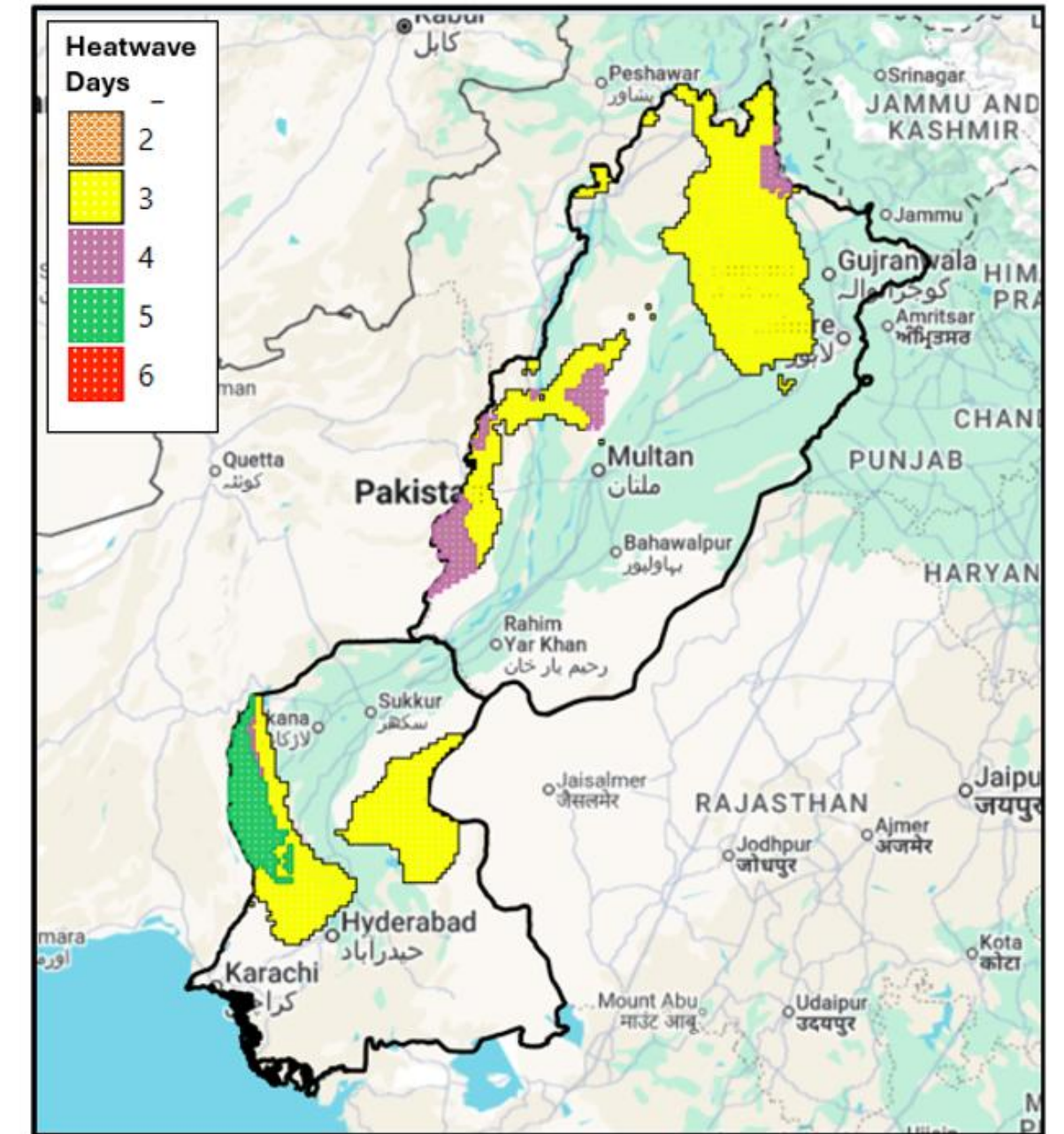
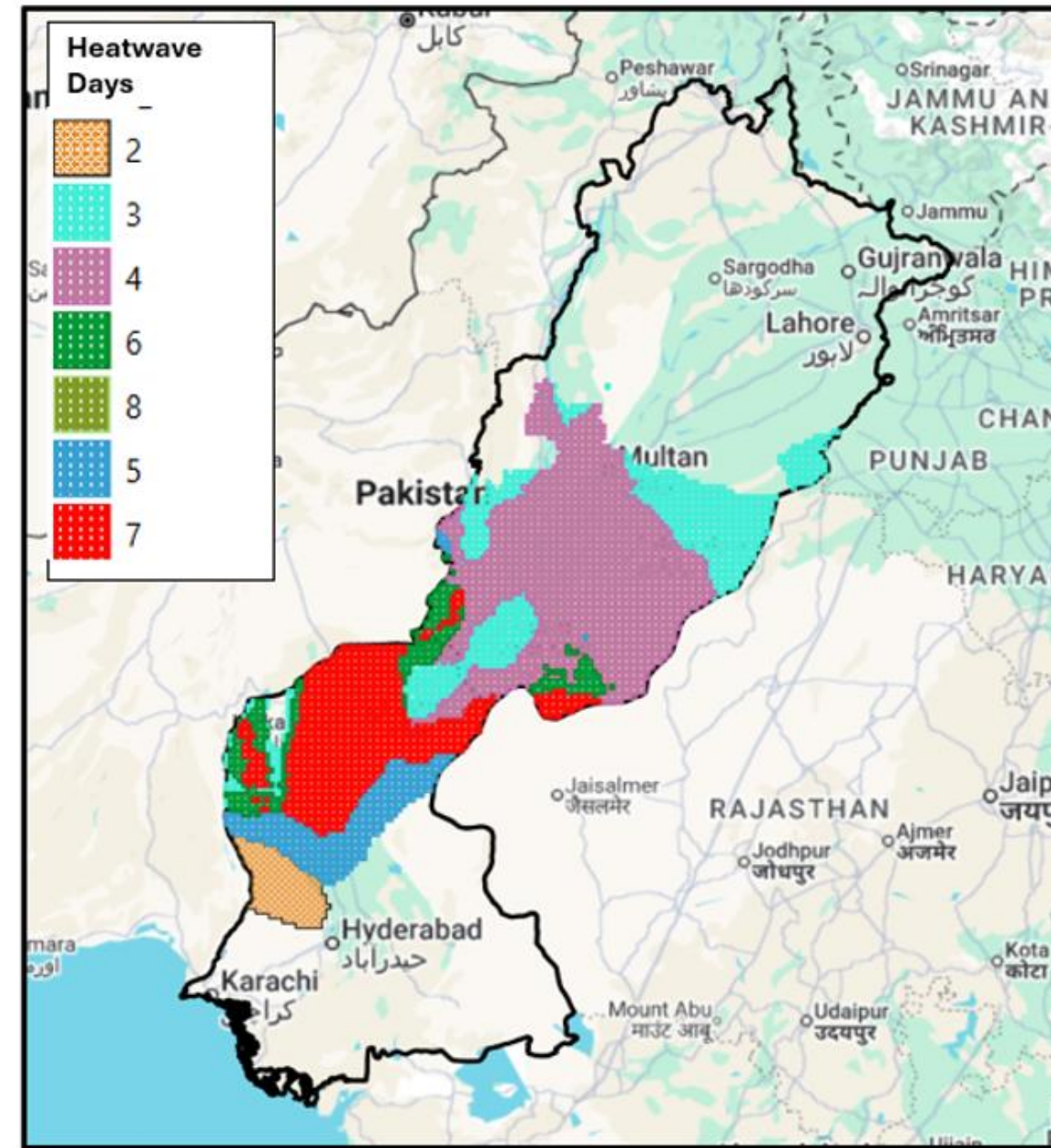
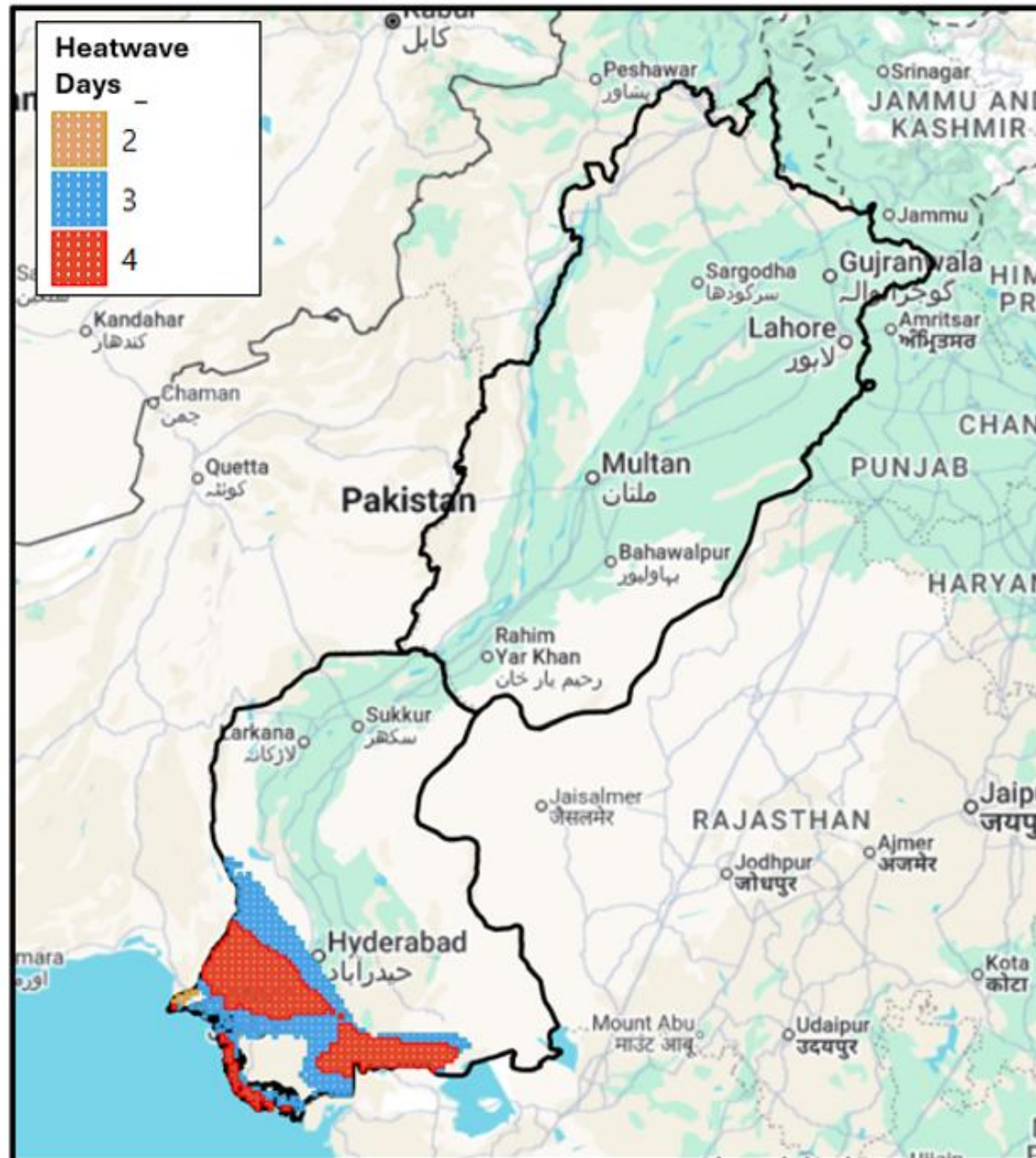
Heatwave Detection and Event Mapping

Heatwave Events, 2026

Event	Event period	Duration
2026-01	1-4 May 2026	4 days

Event	Event period	Duration
2026-02	23-29 May 2026	7 days

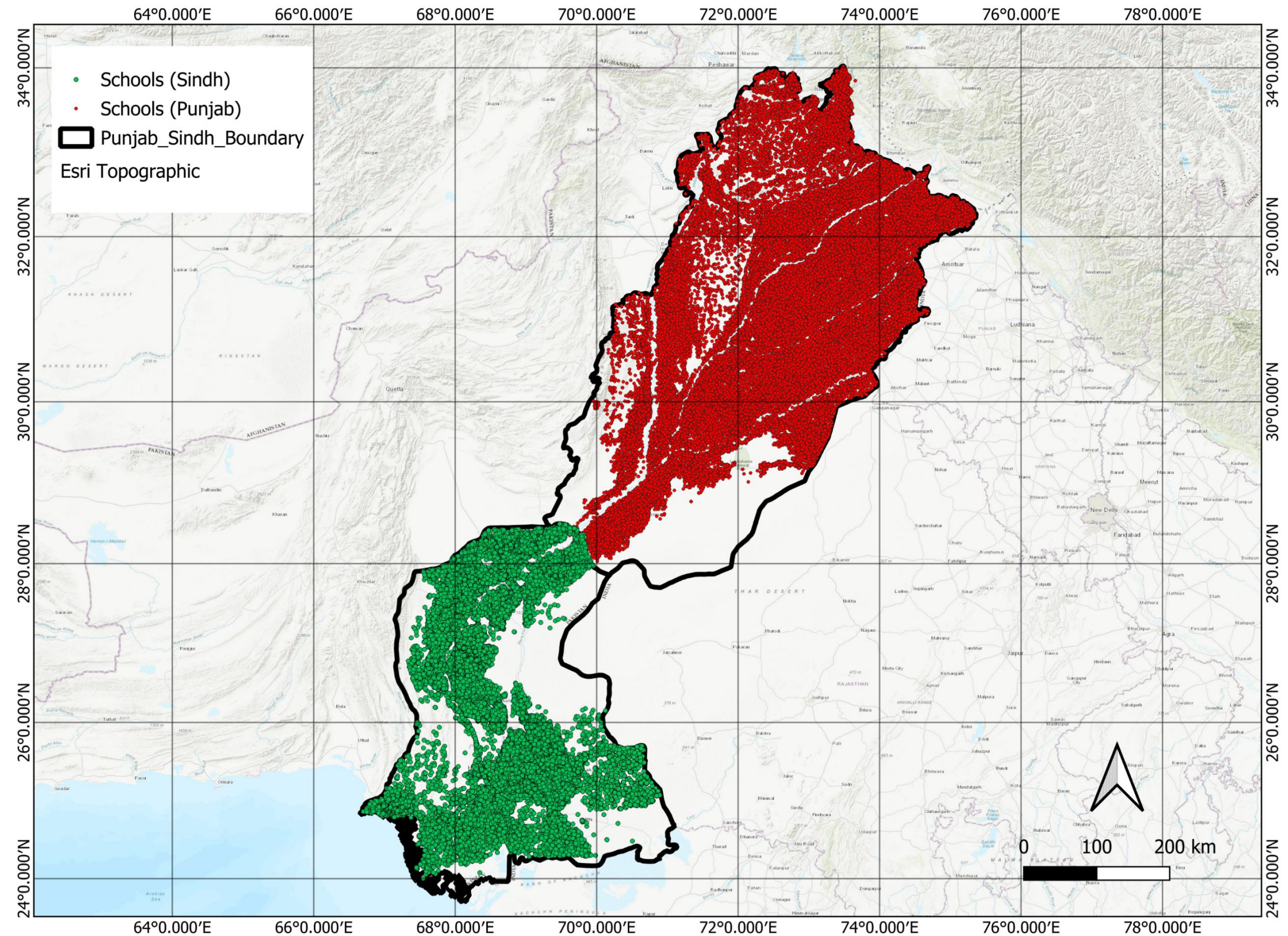
Event	Event period	Duration
2026-03	8-12 June 2026	5 days



Schools Data

School Locations; Punjab and Sindh

- 36756 in Sindh (2025 data)
- 52337 in Punjab (2017 data)



Workstream 1: Climate Extremes & School Exposure Atlas

Policy Question 1: Where and when are schools in Punjab and Sindh exposed to extreme heat, extreme rainfall, and dry spells, and how many students, by school level and gender, are affected?

Approach:

- Percentile-based episode detection on 1 km daily grids (e.g., heatwave = 3+ consecutive days above the local 95th percentile; extreme rainfall above the 99th percentile; floods)
- Episodes localised in space and time at 1 km resolution; can be aggregated to district and tehsil level
- Overlay on geo-located schools to estimate exposed schools and enrolled students
- Disaggregation by province, school level, and gender composition

Deliverable:

Climate extremes and school exposure atlas, with ranked district and school lists for both provinces.

Workstream 2: Enrolment Disruption Analysis (Punjab)

Policy Question 2: Do extreme climate events cause measurable enrolment disruption in Punjab government schools, how large, how persistent, and which school types are most affected?

Approach:

- Event-study design on the five-year monthly enrolment panel: exposed vs. matched unexposed schools around each climate episode
- School-by-calendar-month fixed effects to separate climate effects from strong seasonal enrolment cycles
- Estimates of enrolment drop, recovery duration, and persistent decline, by hazard type
- Heterogeneity analysis: girls' schools, primary schools, urban vs. rural

Deliverable:

Effect estimates by hazard type and school characteristics (quantitative finding).

Caveat:

Enrolment shifts between private and government schools may bias urban estimates; results measure enrolment disruption, not attendance or learning loss.

Workstream 3: Risk Prioritisation & Lahore SES Case Study

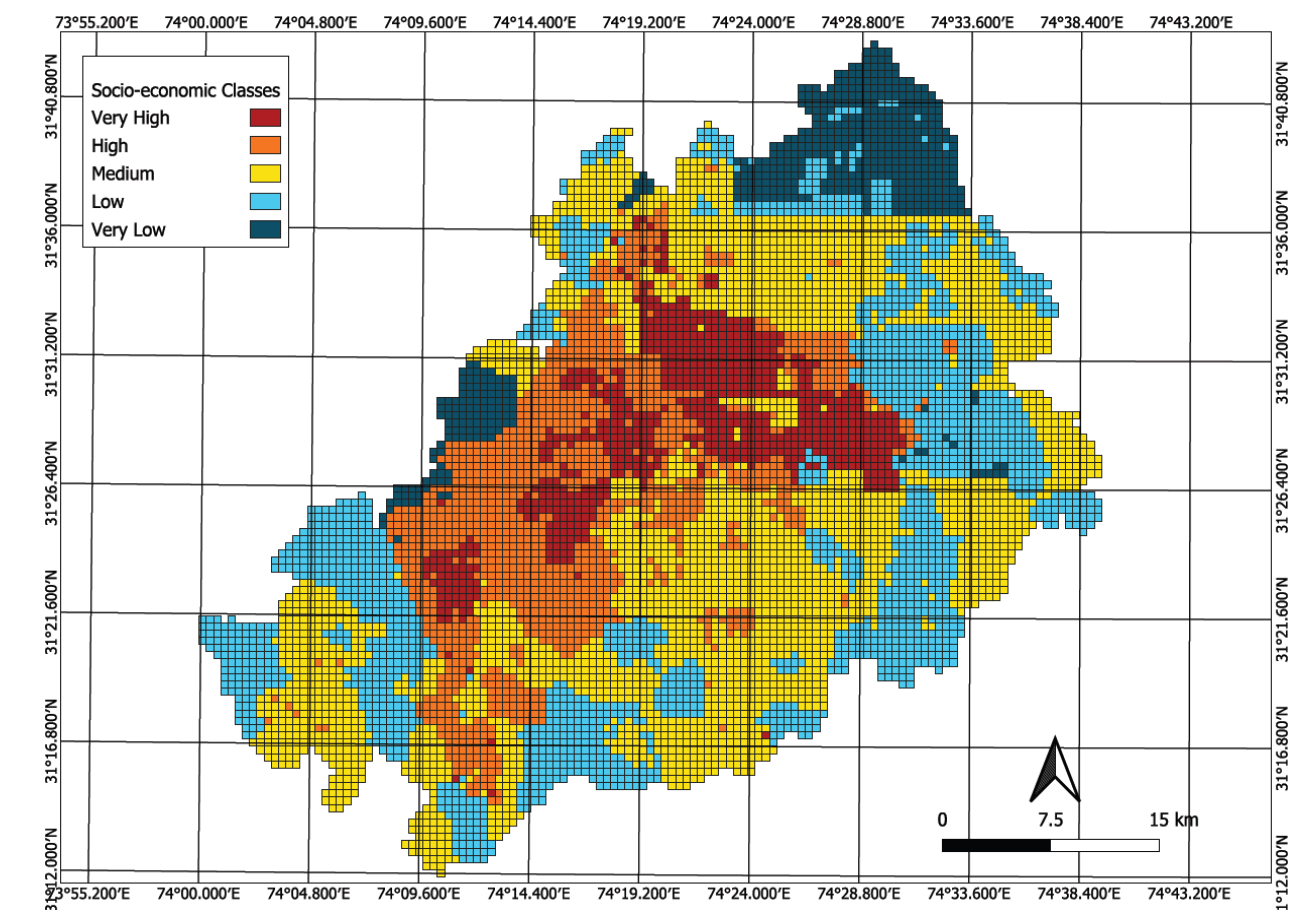
Policy Question 3: Which districts and school categories face compounding climate exposure and educational vulnerability, and should be prioritised for climate-resilience investment? (Time-permitting, future work)

Tier A: Both provinces

- Composite risk index combining exposure (WS1), observed sensitivity (WS2, Punjab), school characteristics, and district-level proxies (population density, poverty estimates)
- Output: ranked priority districts and school categories for resilience planning

Tier B: Lahore nested case study

- 1 km SES map × climate exposure × school locations: identifies high-exposure, low-SES-catchment government schools; smog and heat as featured hazards
- Methodological pilot demonstrating SES-integrated risk mapping; scaling requires SES data for other districts (links to RQ1 data-gap findings)



Lahore government-school catchments by SES class (1 km)

Deliverable:

Priority rankings for provincial education departments; direct input to the policy brief and stakeholder roundtable.

Next Steps and Expected Outcomes

1

May–June
(Completed)

- Completed literature mapping across multiple academic databases and extracted relevant evidence.
- Finalized the inception report.
- Onboarded research team members.
- Initiated Key Informant Interviews (KIIs).
- Extracted education and climate datasets from open-source databases.

2

July
(Ongoing)

- Conducting in-depth analysis of 300+ scoping review papers.
- Completed KIIs in Punjab and Islamabad.
- Initiated KIIs in Sindh, Khyber Pakhtunkhwa (KP), and Balochistan, with completion expected shortly.
- Started secondary data scraping and preliminary analysis of administrative datasets.
- Submitted the mid-point progress report to DARE-RC.

3

Next Steps
(Month 5)

- Synthesize findings from the scoping review and Key Informant Interviews.
- Conduct preliminary analysis of provincial administrative datasets.
- Finalize the policy brief with evidence-based recommendations.
- Organize a stakeholder roundtable to validate findings and discuss policy implications.

4

Expected
Outcomes

- Comprehensive scoping review report on climate-related educational disruptions.
- Policy brief with actionable recommendations for climate-resilient education planning.
- Stakeholder-validated evidence and insights to inform education policy and planning across Pakistan.

Policy Engagement Update

Planned Activity	Status	Progress Update
Identify and map key policy stakeholders	Completed	Key stakeholders identified across education, climate, and development sectors.
Engage stakeholders through Key Informant Interviews (KIIs)	Ongoing	KIIs completed in Punjab and Islamabad; engagement underway in Sindh, KP, and Balochistan.
Build a multi-stakeholder policy engagement panel	Ongoing	Panel members process of invitation on going from government, development partners, and academia.
Stakeholder validation policy roundtable	Planned	To be held following completion of data analysis and synthesis of findings at end of August
Dissemination of policy brief	Planned	Final policy brief to be shared with stakeholders upon project completion.

Proposed Policy Engagement Panel

- Provincial Education Departments (Punjab, Sindh, KP, and Balochistan)
- Ministry/Departments of Climate Change and Environmental Protection
- Development Partners: UNICEF, World Bank, Asian Development Bank (ADB)
- Researchers and academics working on climate change, education, and resilience
- Key Informants who participated in interviews and wish to remain engaged

Expected Outcome

A collaborative policy dialogue that strengthens the relevance, uptake, and application of the project's findings to support climate-resilient education planning in Pakistan.

Risks and Mitigation Matrix

Challenges/Risks	Mitigation Strategies	Implications
Administrative data on climate and education depends on the openness and data-sharing practices of federal and provincial departments.	Engage data custodians early; triangulate with KII and scoping review evidence where administrative data is incomplete.	Gaps could limit the depth of quantitative analysis in the secondary data phase.
Variable quality and consistency of administrative datasets across institutions.	Apply data quality checks; supplement with satellite-derived and reanalysis climate data (ERA5, CHIRPS).	May narrow the scope of predictive modelling to descriptive analysis only.
<i>KII outside Punjab is difficult to contact, biggest challenge has occurred in Sindh region</i>	<i>DRAE-RC and other referral support from personal connections have been utilized to smooth the process and build and developed trust</i>	<i>Not able to do more comprehensive coverage as in-case of Punjab</i>

Budget Report

A matrix that indicates whether expenses are on track, below budget, or above budget, with an explanation of the variance. Please update the below table:

Approved Budget (PKR)	Expenditure incurred (PKR)	Variance (PKR)	Reasons for over or under spent
Human Resources	7,010,000 (Committed)	0	All HR team members were hired on time and the project is on budget, on time, and in full. Amount of 5,110,000 PKR for Pls will be paid at the conclusion of the project.
Project Cost	1,095,500 (Committed)	0	To be utilized in PE activities
Overhead Cost	810,500	0	Charged by LUMS Office of Research
Sales Tax on Service (budget vs charged by OPM)	1,426,480 (budgeted)	0	As per actual

Updated Work Plan

July 2026

- Complete all Key Informant Interviews (KIIs) across Sindh, KP, and Balochistan
- Complete secondary data collection from provincial education departments and climate-related agencies by end of July

August 2026

- Complete analysis of the scoping review (300+ papers)
 - Complete analysis of KII findings
 - Conduct analysis of education and climate datasets
 - Hold the Policy Dialogue / Stakeholder Roundtable
- Prepare the final report and policy brief

Expected Deliverables

01

Stakeholder Policy Dialogue

A validation roundtable bringing together government, development partners, and academia.

02

Comprehensive Report

Integrating the scoping review, KII findings, and secondary data analysis into one synthesis.

03

Policy Brief/ Blog

Actionable recommendations for climate-resilient education planning in Pakistan.

Thank You

Follow our research journey, policy insights, and events

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